

Comunicazioni Numeriche

Prof. Luca Sanguinetti (6cfu), Prof. Nicola Andriolli(3cfu)

About me

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Syllabus

SANGUINETTI	ANDRIOLLI
Fourier analysis of analogue signals	Elements of probability
Linear systems and filters	Random variables
Sampling theorem	Stochastic processes
Coding Theory: Block codes	
Base-band digital communication systems	
Pass-band digital communication systems	

Class schedule

lun 2/3	mar 3/3	mer 4/3	gio 5/3	ven 6/3
	08:30 - 11:30 COMUNICAZIONI NUMERICHE INGEGNERIA INFORMATICA (3) Ing A26 N. ANDRIOLLI,L. SANGUINETTI		08:30 - 11:30 PROGETTAZIONE DI RETI INFORMATICHE INGEGNERIA INFORMATICA (3) Ing SI7 E. MINGOZZI	
	11:30 - 13:30 ELETTRONICA DIGITALE INGEGNERIA INFORMATICA (3) Ing A26 M. PIOTTO,M. DEI			
12:30 - 14:30 ELETTRONICA DIGITALE INGEGNERIA INFORMATICA (3) Ing C11 M. PIOTTO,M. DEI			12:30 - 14:30 ELETTRONICA DIGITALE INGEGNERIA INFORMATICA (3) Ing SI1 M. PIOTTO,M. DEI	
14:30 - 16:30 PROGETTAZIONE DI RETI INFORMATICHE INGEGNERIA INFORMATICA (3) Ing SI5 E. MINGOZZI	14:30 - 17:30 LABORATORIO DI INFORMATICA APPLICATA INGEGNERIA INFORMATICA (3) Ing SI1 T. URICCHIO		14:30 - 17:30 COMUNICAZIONI NUMERICHE INGEGNERIA INFORMATICA (3) Ing A26 N. ANDRIOLLI,L. SANGUINETTI	13:30 - 15:30 ELETTRONICA DIGITALE INGEGNERIA INFORMATICA (3) Ing B32 M. PIOTTO,M. DEI
16:30 - 18:30 LABORATORIO DI INFORMATICA APPLICATA INGEGNERIA INFORMATICA (3) Ing SI5 T. URICCHIO			15:30 - 17:30 COMUNICAZIONI NUMERICHE INGEGNERIA INFORMATICA (3) Ing B32 N. ANDRIOLLI,L. SANGUINETTI	

Channel Calendar (TEAMS)

Monday

Tuesday

Wednesday

Thursday

Friday



About course and exam

Structure

- Lectures
- Excersises
- Matlab scripts

Repository

- Teams

Exam

- Written
- Oral

Materials

- Lecture notes
- Slides
- Textbooks
 - M. Luise, G. Vitetta, Teoria dei Segnali, McGraw Hill.
 - S. Haykin, [Digital Communication Systems](#), Wiley.

Introduction

Hystorical Background

Information Theory and Coding

- Information Theory (Claude Shannon, 1948)
- Coding Theory (Golay, 1949, Hamming, 1950, Turbo Codes, 1993, LDPC, 1962)

The Internet

- ARPANET, 1971
- Turning point: World Wide Web (Berners-Lee, 1990), commercialized in 1994

Wireless Communications

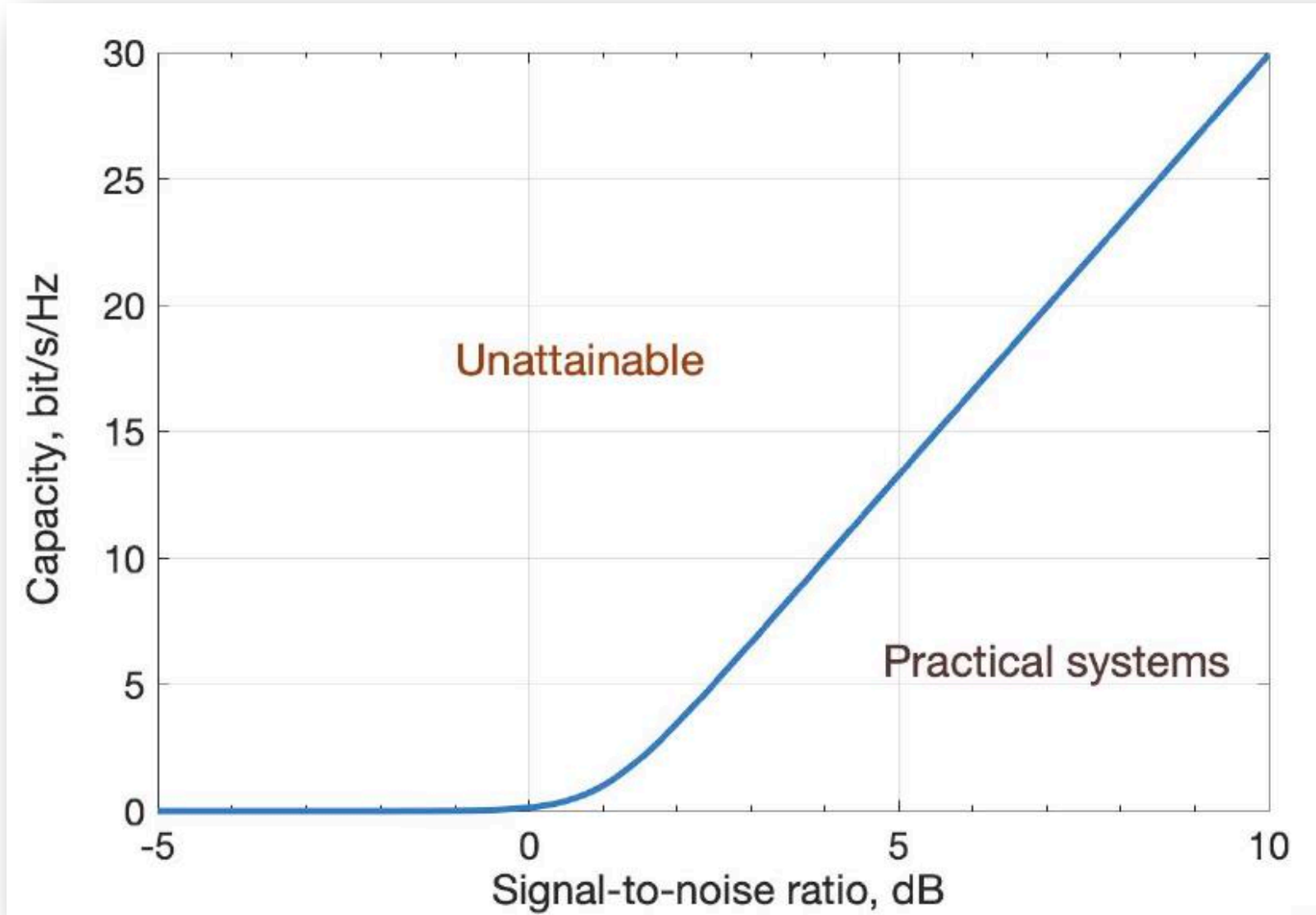
- Electromagnetic theory of light (Maxwell, 1864)
- Radio experiments (Guglielmo Marconi, 1900)
- GSM (2G, 1988), UMTS (3G, 2000), LTE (4G, 2010)
- 5G (2020), 6G (2030)

Information Theory

- Claude Shannon ``A Mathematical Theory of Communication'', 1948
 - Prior to Shannon's 1948 paper, it was believed that increasing the rate of transmission over a channel would increase the probability of error
 - By surprise, Shannon proved that this was not true, provided the transmission rate was below the channel capacity
 - The channel capacity is defined as the maximum rate of reliable (error-free) information transmission through the channel

$$\text{Capacity} = \log_2 \left(1 + \frac{\text{Signal}}{\text{Noise}} \right) \quad \text{bit/s/Hz}$$

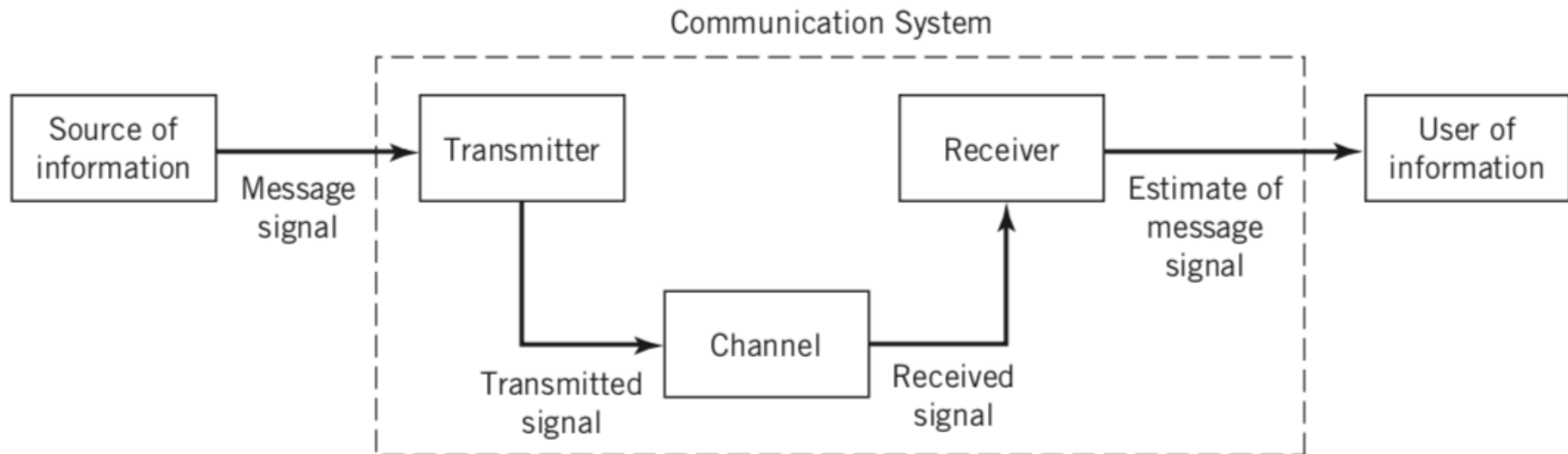
Channel Capacity



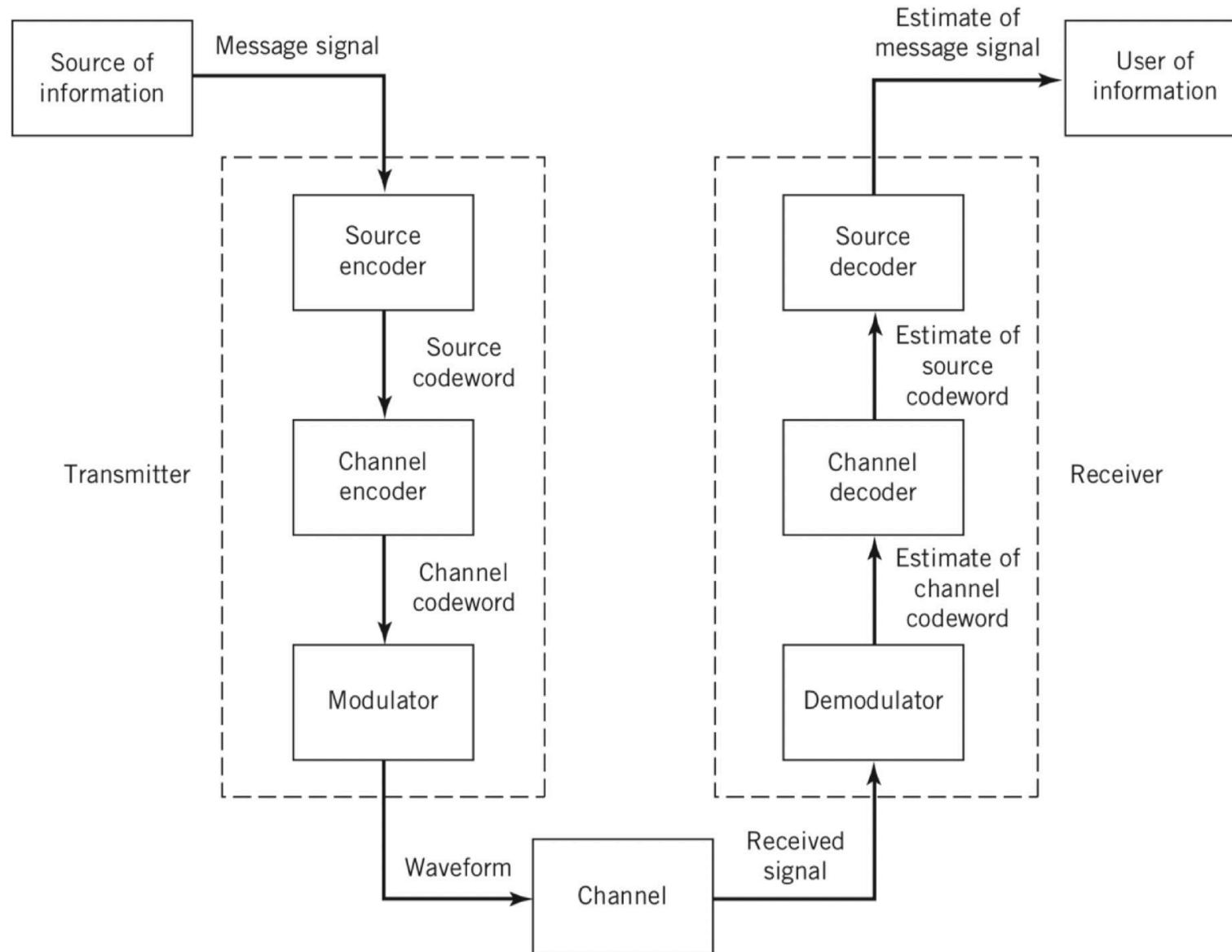
Elements of a Communication System

Study of communication systems involves:

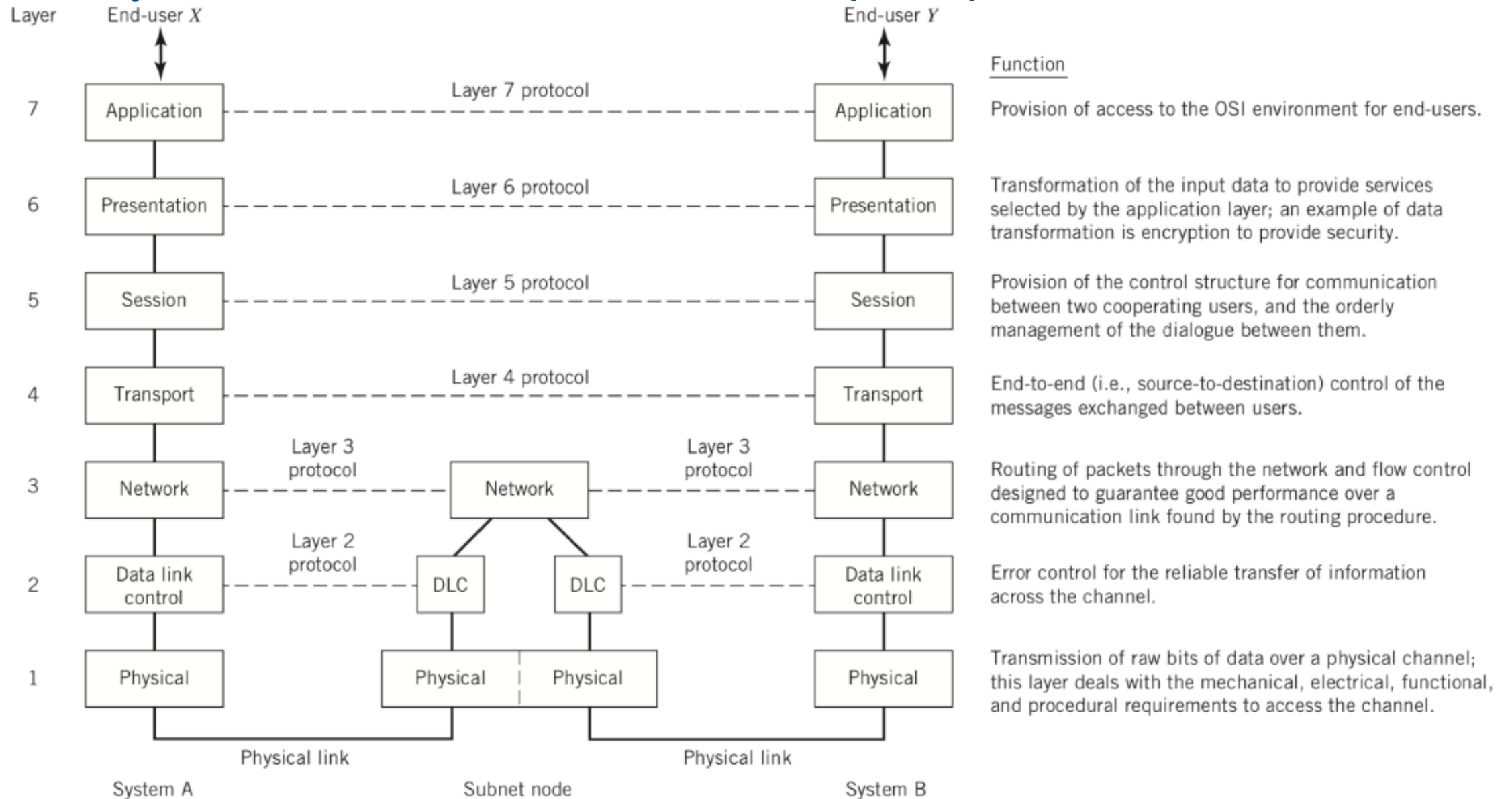
- the processing of a modulated signal generated at the transmitter
 - to facilitate its transportation across a physical channel;
- subsequent processing of the received signal in the receiver
 - to deliver an estimate of the original signal.



Digital Communication System



Open System Interconnection (OSI) Model



Digital Communications

- Physical layer - Lowest layer of the OSI model
 - Communication accomplished by means of *modulation* in the transmitter, transmission across the channel, and *demodulation* in the receiver (**modem**)
- Data-link Layer
 - Error correction or detection, packet retransmission
 - Responsible for sharing the transmission medium; medium access control (MAC) sublayer
- Network layer
 - Routing of information, quality of service, flow control (to ensure that the network does not become congested).

Networks

A communication network consists of an interconnection of nodes

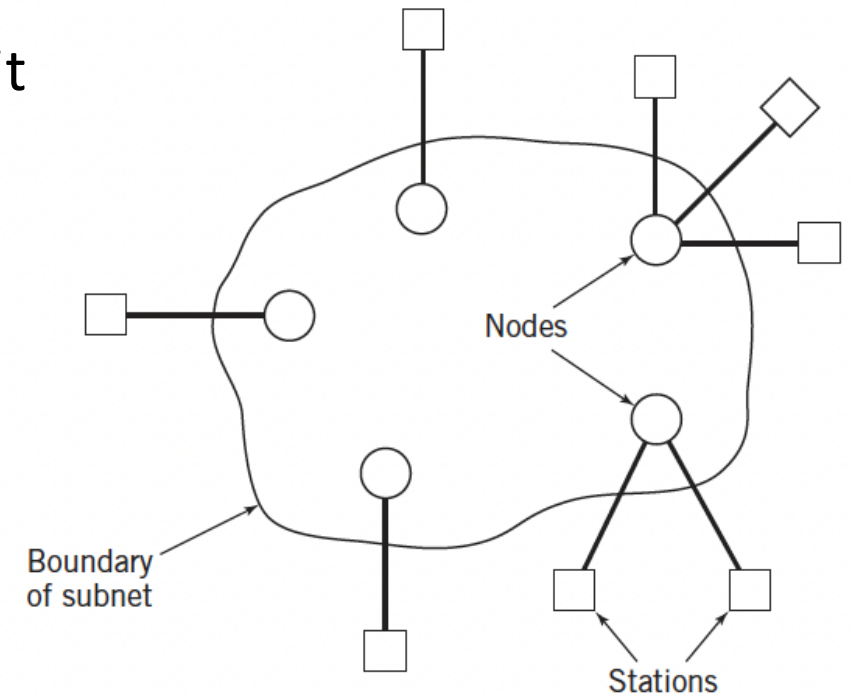
- Primary purpose of these nodes is to route data through the network
- Each node has one or more stations attached to it

Circuit switching (telephone)

- Connected link sequence, controlled by centralized unit
- Exclusive use of two stations wishing to communicate

Packet switching

- Sharing is done on a demand basis
- Store and forward approach



Basic Modes of Communications

Broadcasting (Radio, TV)

- A single powerful transmitter and numerous receivers, relatively inexpensive to build.

Point-to-point communication

- Communication takes place over a link between a single transmitter and a receiver.

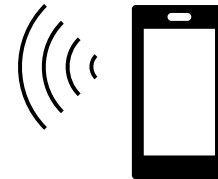
Point-to-multipoint communication (Cellular, WiFi)

- Communication takes place over a link between a single transmitter and multiple receivers.

Multiple Access Techniques

Multiple access

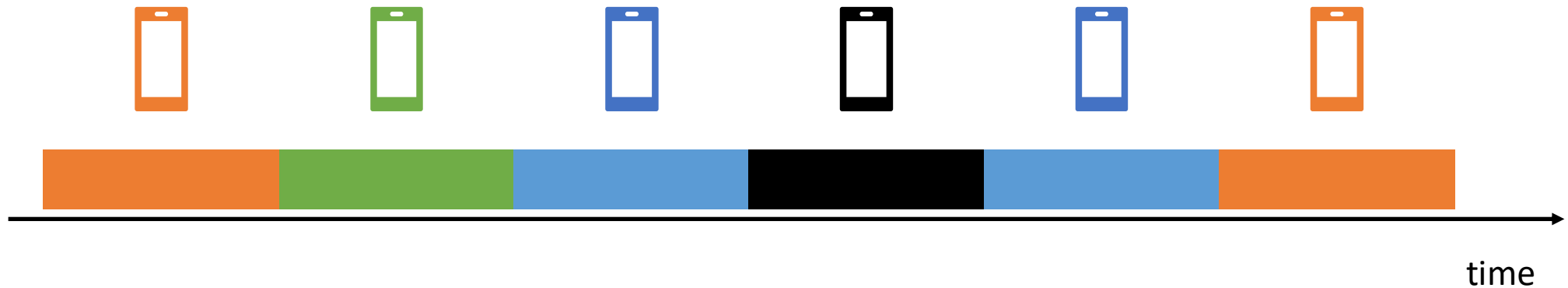
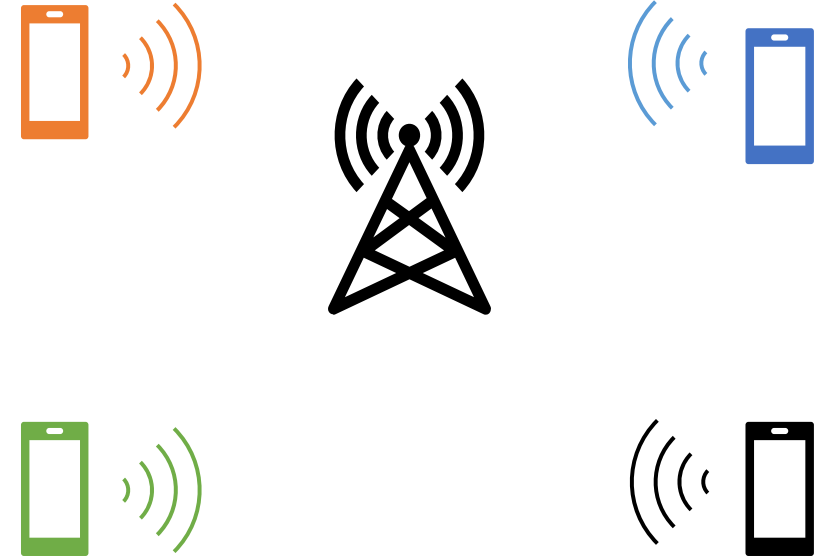
- Sharing of a communication channel



Multiple Access Techniques

Time-division multiple access (TDMA)

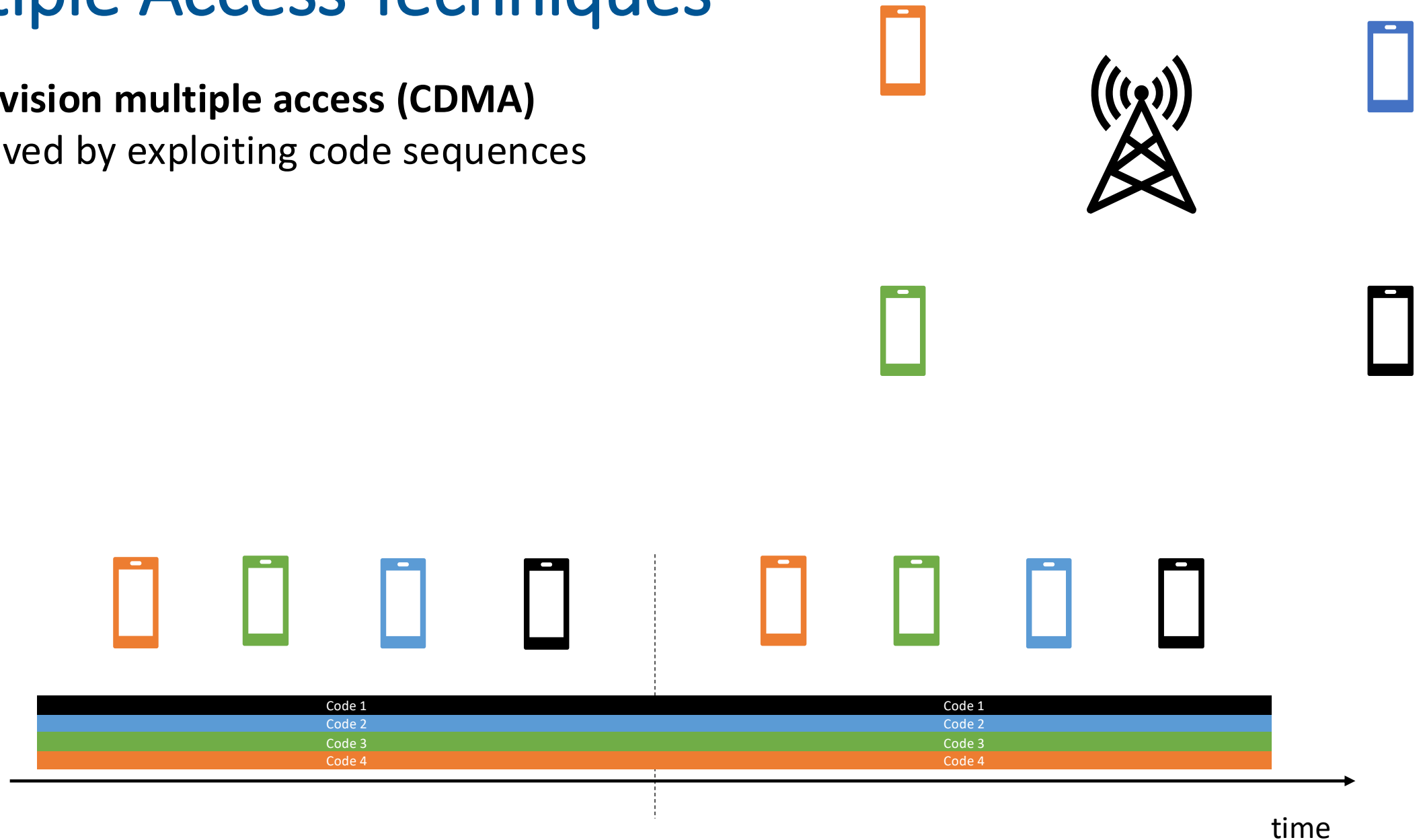
- Disjoint time intervals used for different terminals



Multiple Access Techniques

Code-division multiple access (CDMA)

- achieved by exploiting code sequences



Multiple Access Techniques

Time-division multiple access (TDMA)

- Disjoint time intervals used for different terminals

Code-division multiple access (CDMA)

- Resource allocation is achieved by exploiting code sequences

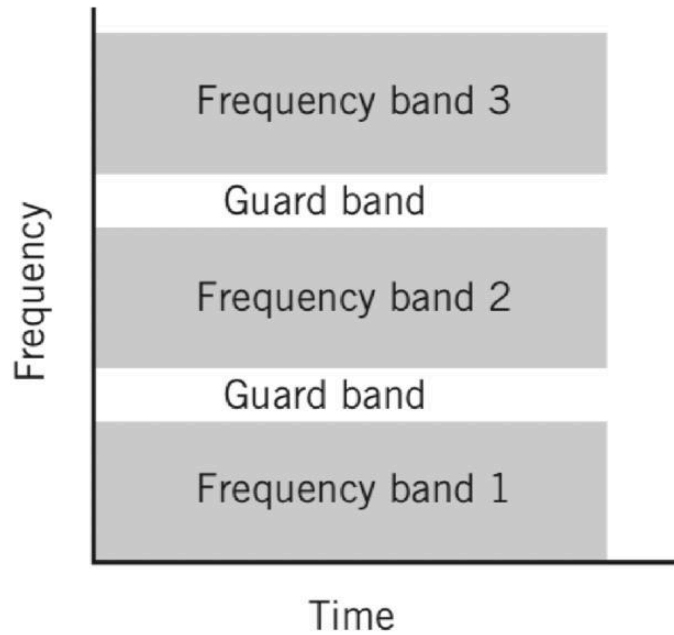
Frequency-division multiple access (FDMA)

- Disjoint subbands of frequency used for different terminals

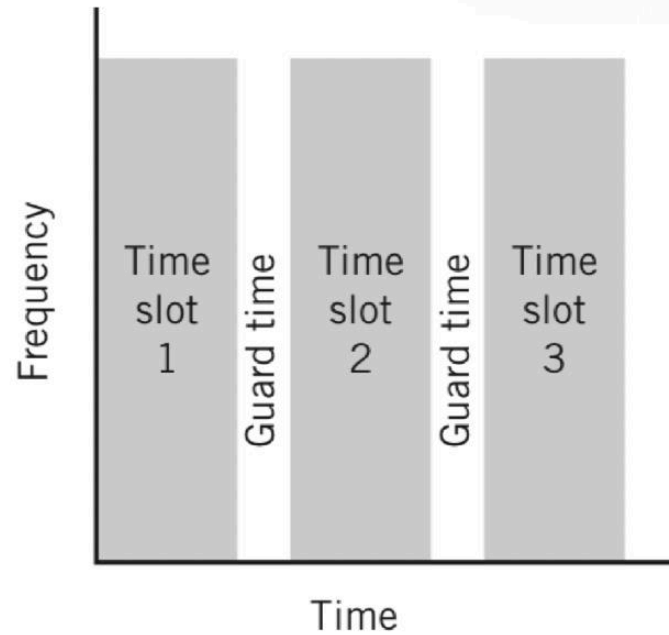
Space-division multiple access (SDMA)

- Resource allocation is achieved by exploiting the spatial separation of users

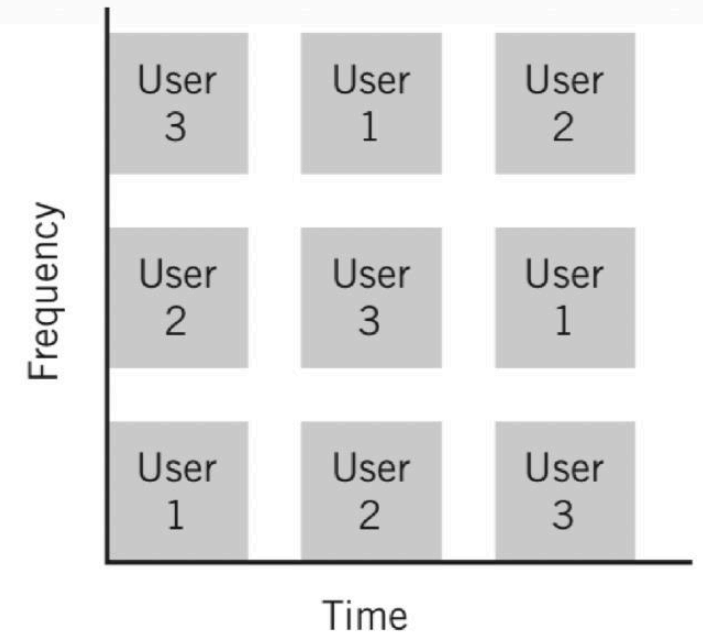
FDMA and TDMA



FDMA



TDMA



Hybrid FDMA-TDMA

Radio Frequency Spectrum

Range of electromagnetic frequencies used for wireless communication.

- It covers frequencies from 3 Hz to 300 GHz.

Key Uses of the Radio Spectrum:

- Broadcasting (AM/FM radio, TV signals)
- Mobile Networks (3G, 4G, 5G)
- Wi-Fi & Bluetooth (Wireless data transfer)
- Satellite Communication (GPS, TV, weather)
- Radar Systems (Aviation, military, weather tracking)

Spectrum Management:

Managed by regulatory bodies (e.g., FCC in USA, ITU globally)

Radio Frequency Spectrum: Ranges

DESIGNATION	ABBREVIATION	FREQUENCIES	FREE-SPACE WAVELENGTHS
Very low frequency	VLF	3 kHz to 30 kHz	100 km to 10 km
Low frequency	LF	30 kHz to 300 kHz	10 km to 1 km
Medium frequency	MF	300 kHz to 3 MHz	1 km to 100 m
High frequency	HF	3 MHz to 30 MHz	100 m to 10 m
Very high frequency	VHF	30 MHz to 300 MHz	10 m to 1 m
Ultrahigh frequency	UHF	300 MHz to 3 GHz	1 m to 100 mm
Super-high frequency	SHF	3 GHz to 30 GHz	100 mm to 10 mm
Extremely high frequency	EHF	30 GHz to 300 GHz	10 mm to 1 mm

UNITED STATES FREQUENCY ALLOCATIONS

RADIO SERVICES COLOR LEGEND

 AERONAUTICAL MOBILE	 INTER-SATELLITE	 RADIO ASTRONOMY
 AERONAUTICAL MOBILE SATELLITE	 LAND MOBILE	 RADIO DETERMINATION SATELLITE
 AERONAUTICAL RADIO NAVIGATION	 LAND MOBILE SATELLITE	 RADIO LOCATION
 AMATEUR	 MARITIME MOBILE	 RADIO LOCATION SATELLITE
 AMATEUR SATELLITE	 MARITIME MOBILE SATELLITE	 RADIO NAVIGATION
 BROADCASTING	 MARITIME RADIO NAVIGATION	 RADIO NAVIGATION SATELLITE
 BROADCASTING SATELLITE	 METEOROLOGICAL	 SPACE OPERATION
 EARTH EXPLORATION SATELLITE	 METEOROLOGICAL SATELLITE	 SPACE RESEARCH
 FIXED	 MOBILE	 STANDARD FREQUENCY AND TIME SIGNAL
 FIXED SATELLITE	 MOBILE SATELLITE	 STANDARD FREQUENCY AND TIME SIGNAL SATELLITE

ACTIVITY CODE

 FEDERAL EXCLUSIVE	 FEDERAL NON-FEDERAL SHARED
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SERVICE	EXAMPLE	DESCRIPTION
Primary	FIXED	Capital Letters
Secondary	Mobile	1st Capital with lower case letters

 U.S. DEPARTMENT OF COMMERCE
National Telecommunications and Information Administration
Office of Spectrum Management
JANUARY 2016



Numeri Complessi

Rappresentazione cartesiana

Unità immaginaria $i^2 = -1$

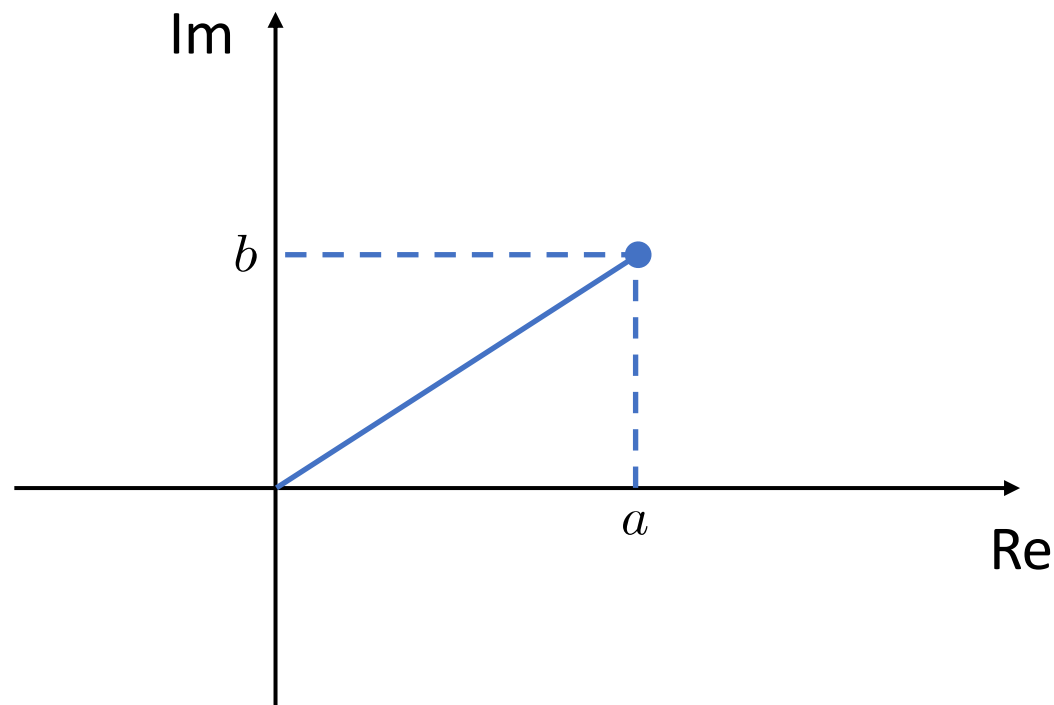
$$z = a + ib$$

Parte reale

Parte Immaginaria

$$a = \operatorname{Re}\{z\}$$

$$b = \operatorname{Im}\{z\}$$

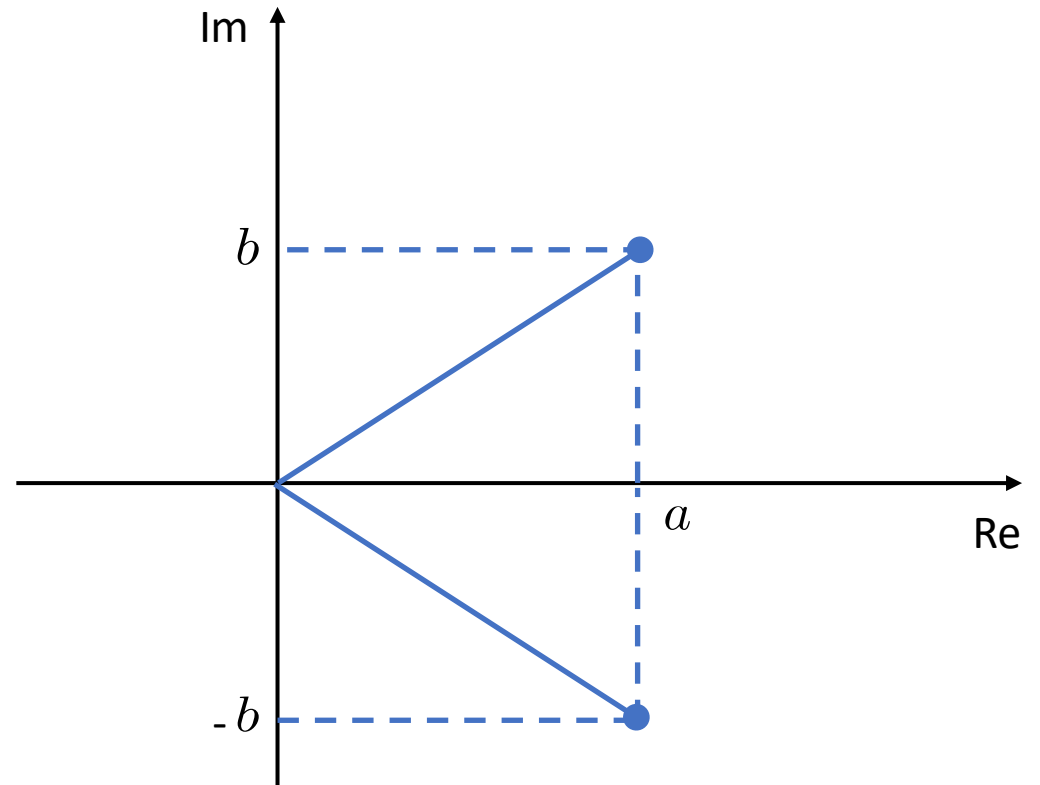


Coniugato

$$z^* = a - ib$$

$$a = \operatorname{Re}\{z\} = \frac{z+z^*}{2}$$

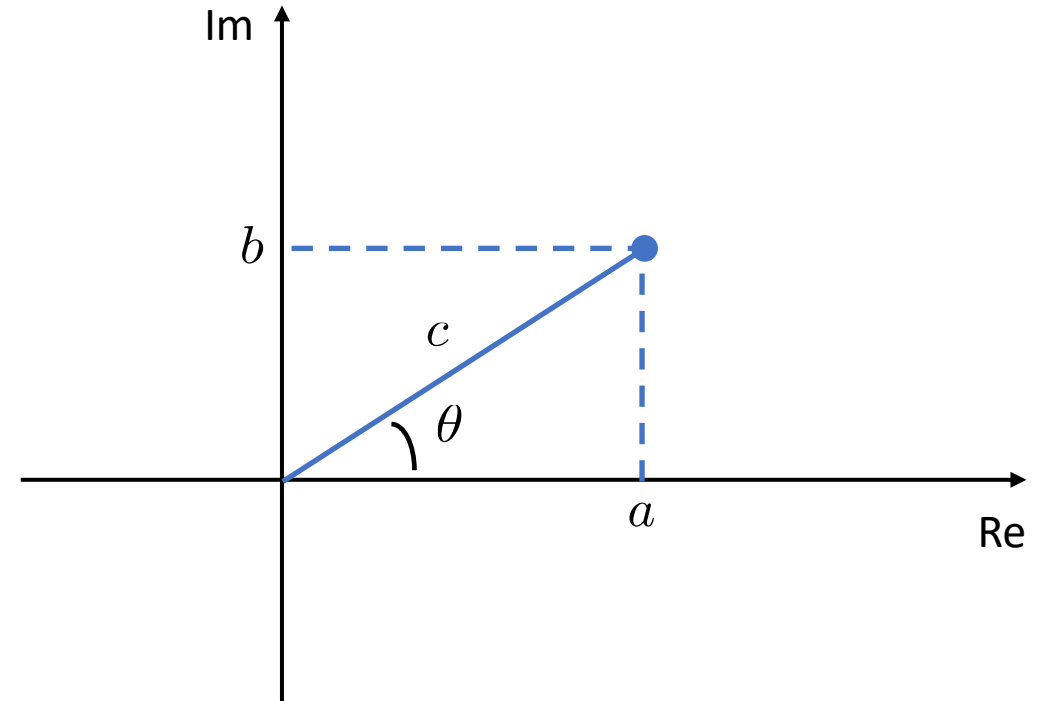
$$b = \operatorname{Im}\{z\} = \frac{z-z^*}{2i}$$



Rappresentazione Polare

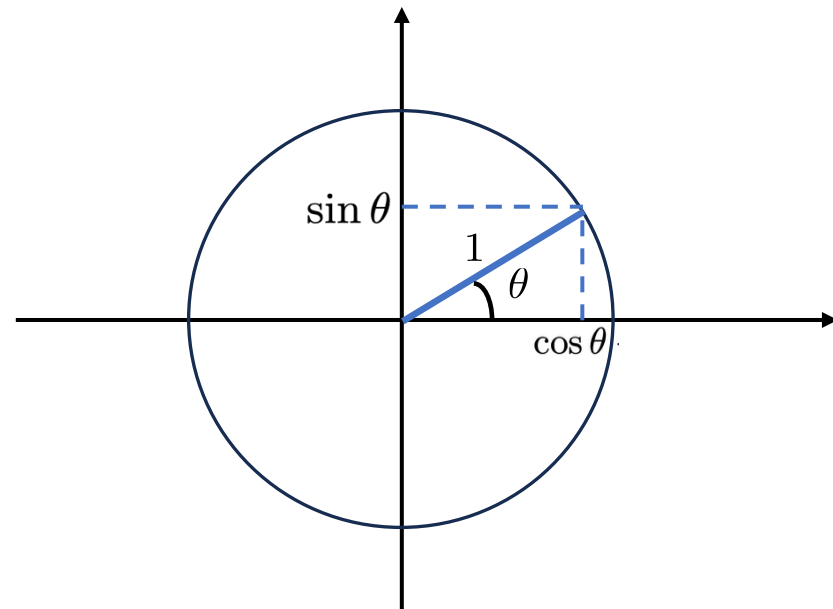
$$z = \underset{\substack{\uparrow \\ \text{Modulo}}}{c} e^{i\theta} \quad \leftarrow \text{Fase} \quad \left\{ \begin{array}{l} c = \sqrt{a^2 + b^2} \\ \theta = \operatorname{tg}^{-1} \frac{b}{a} \end{array} \right.$$

$$z^* = a - ib = c_1 e^{-i\theta_1}$$



Trasformazione Polare/Cartesiana

$$e^{i\theta} = \cos \theta + i \sin \theta$$



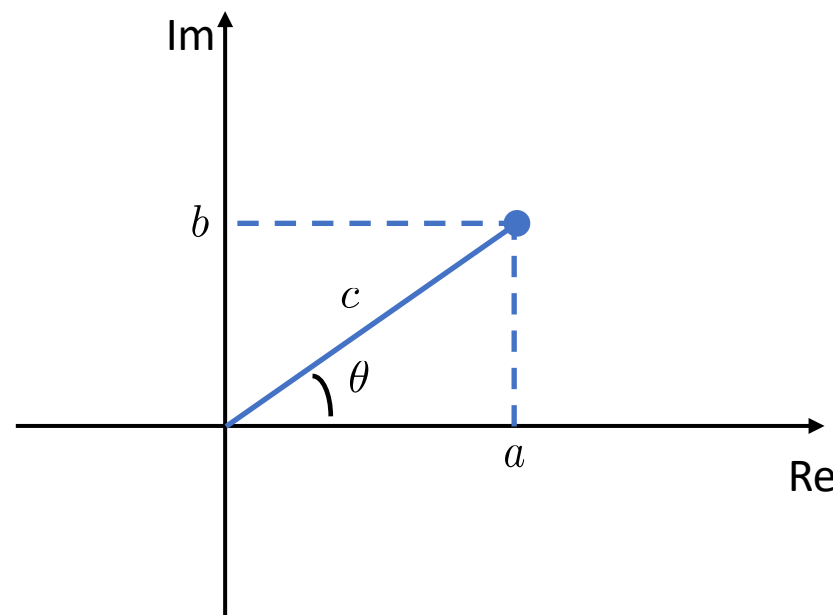
$$ce^{i\theta} = c \cos \theta + ci \sin \theta = a + ib$$

$$a = c \cos \theta$$

$$b = c \sin \theta$$

$$c = \sqrt{a^2 + b^2}$$

$$\theta = \operatorname{tg}^{-1} \frac{b}{a}$$



Operazioni Algebriche: Somma/Differenza

$$z_1 = a_1 + ib_1 = c_1 e^{i\theta_1}$$

$$z_2 = a_2 + ib_2 = c_2 e^{i\theta_2}$$

$$z_1 + z_2 = a_1 + a_2 + i(b_1 + b_2)$$

$$z_1 - z_2 = a_1 - a_2 + i(b_1 - b_2)$$

$$z_1 \pm z_2 = a_1 \pm a_2 + i(b_1 \pm b_2)$$

Operazioni Algebriche: Prodotto

$$\begin{aligned} z_1 \cdot z_2 &= (a_1 + ib_1)(a_2 + ib_2) \\ &= a_1a_2 + ia_1b_2 + ia_2b_1 - b_1b_2 \\ &= (a_1a_2 - b_1b_2) + i(a_1b_2 + a_2b_1) \end{aligned}$$

$$\begin{aligned} z_1 \cdot z_2 &= c_1 e^{i\theta_1} \cdot c_2 e^{i\theta_2} \\ &= c_1 c_2 e^{i(\theta_1 + \theta_2)} \end{aligned}$$

Operazioni Algebriche: Divisione

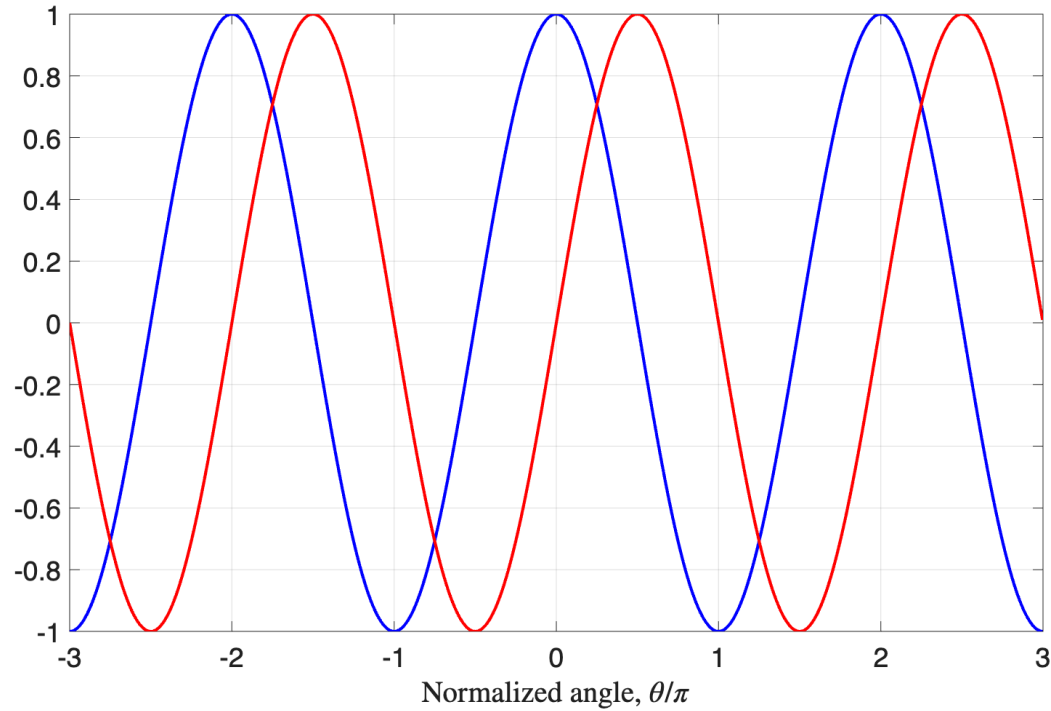
$$\begin{aligned}\frac{z_1}{z_2} &= \frac{z_1}{z_2} \frac{z_2^*}{z_2^*} = \frac{z_1 z_2^*}{|z_2|^2} \\ &= \frac{a_1 a_2 + b_1 b_2 + i(a_2 b_1 - a_1 b_2)}{a_2^2 + b_2^2}\end{aligned}$$

$$\begin{aligned}\frac{z_1}{z_2} &= z_1 \frac{1}{z_2} \\ &= \frac{c_1}{c_2} e^{i(\theta_1 - \theta_2)}\end{aligned}$$

$$\frac{1}{z_2} = \frac{z_2^*}{|z_2|^2} = \frac{1}{c_2} e^{-i\theta_2}$$

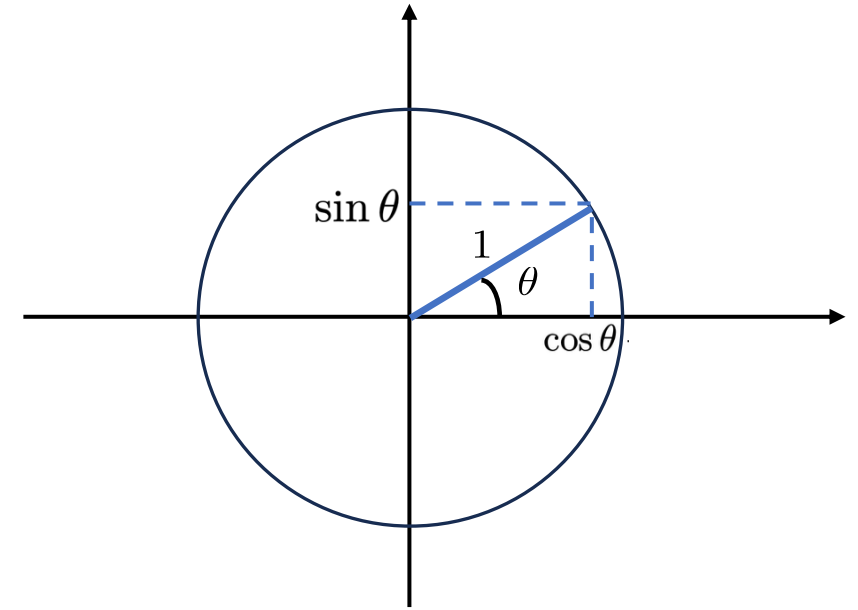
Funzioni seno e coseno

Funzioni seno e coseno



$$\cos(\theta) = \cos(-\theta)$$

$$\sin(\theta) = -\sin(-\theta)$$



$$\cos^2(\theta) + \sin^2(\theta) = 1$$

Funzioni seno e coseno

$$\cos(\alpha \pm \beta) = \cos(\alpha) \cos(\beta) \mp \sin(\beta) \sin(\alpha)$$

$$\sin(\alpha \pm \beta) = \sin(\alpha) \cos(\beta) \pm \cos(\alpha) \sin(\beta)$$

$$\cos(2\alpha) = \cos^2(\alpha) - \sin^2(\alpha)$$

$$\cos^2(\alpha) = \frac{1 + \cos(2\alpha)}{2}$$

$$\sin(2\alpha) = 2 \sin(\alpha) \cos(\alpha)$$

$$\cos(\theta) = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$\sin(\theta) = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$